

CHRONO RIFT

Turnaround Time Analysis

CS 2006 — Operating Systems | Spring 2026

1. Scheduling Model Overview

Chrono Rift uses a discrete turn-based scheduler built on a stamina economy. Unlike continuous-time schedulers (e.g., Linux CFS), there is no wall-clock tick; instead the arbiter advances a turn queue and replenishes stamina on each call to `pick_next_actor()`. Every entity receives +30 stamina when selected. Turnaround time (TAT) is therefore expressed in turns, with a lightweight scheduling overhead (~5 ms per arbiter loop) added for real-time context.

The key scheduler constants extracted from `arbiter.cpp`:

Constant	Value	Meaning
STA_REGEN	30	Stamina gained per turn when the entity is selected by the scheduler
STA_SKIP_REGEN	40	Stamina gained when SKIP action is forced (low-stamina auto-rest)
STA_MIN_ACT	10	Minimum stamina required to act; entities below this are force-skipped
STA_STRIKE	25	Stamina cost of a STRIKE action
STA_EXHAUST	20	Stamina cost of an EXHAUST action
STA_WEAPON	30	Stamina cost of USE_WEAPON action
STA_SWAP	10	Stamina cost of SWAP_IN action
STA_HEAL	15	Stamina cost of HEAL action
STA_ULTIMATE	50	Stamina cost of ULTIMATE (also sets actor stamina to 0)
STA_ACQUIRE	10	Stamina cost of ACQUIRE_ARTIFACT action
NPC_TIMEOUT_SEC	3	Seconds before an NPC turn auto-skips (if <code>action_ready_cond</code> not signalled)
STUN_DURATION_SEC	3	Default per-entity stun duration in seconds
ULTIMATE_PAUSE_SEC	10	Enemy stun duration triggered by the player's Ultimate Ability

2. Turnaround Time Definition & Formula

Turnaround Time (TAT) for a single action cycle is the interval from the moment the arbiter assigns the entity its turn to the moment the arbiter finishes applying the entity's chosen action and the next turn assignment begins. In the discrete model:

$$\text{TAT_turns}(\text{S_before}, \text{cost}) = \text{ceil} \left(\frac{\text{MAX_STA} - \max(0, \text{S_before} - \text{cost})}{\text{STA_REGEN}} \right)$$

Where:

- **S_before** = stamina when the entity's turn starts (after +STA_REGEN is added by the scheduler)
- **cost** = stamina deducted by the chosen action
- **MAX_STA** = max_stamina for this entity type (100 for players, 150 for enemies)
- **TAT_turns** = how many turns must elapse before this entity acts again

Intuition: a higher-cost action leaves the entity with less stamina, so the scheduler needs more regen cycles to bring it back to MAX_STA — meaning a longer wait before the next turn.

Converting to wall time (approximate, adds one scheduling overhead per turn):

$$\text{TAT_sec} \approx \text{TAT_turns} \times \text{SCHED_OVERHEAD} \quad (\text{SCHED_OVERHEAD} \approx 5 \text{ ms per arbiter loop})$$

For the Ultimate Ability, the entity's stamina is explicitly set to 0 after the action, regardless of S_before. The formula simplifies to:

$$\text{TAT_turns}(\text{ULTIMATE}) = \text{ceil} \left(\frac{\text{MAX_STA}}{\text{STA_REGEN}} \right) = \text{ceil}(100/30) = 4 \text{ [player]}$$

3. Per-Action Turnaround Time Analysis

The table below shows TAT for each action type under a representative scenario: entity starts its turn at MAX_STA (full stamina) because the scheduler just added STA_REGEN. For players MAX_STA = 100; for enemies MAX_STA = 150.

Action	STA Cost	STA After (Player)	TAT (turns) Player	TAT (turns) Enemy
STRIKE	25	75	1 turn	1 turn
EXHAUST	20	80	1 turn	1 turn
USE_WEAPON	30	70	1 turn	1 turn
SWAP_IN	10	90	1 turn	1 turn
HEAL	15	85	1 turn	1 turn
ACQUIRE_ARTIFACT	10	90	1 turn	1 turn
SKIP (voluntary)	0 (rest)	100	1 turn	1 turn
SKIP (low STA)	0 (rest)	100	1 turn	1 turn
ULTIMATE	50	0	4 turns	5 turns

Key insight: STRIKE (25 cost) → 1 turn TAT for players (stamina goes from 100→75, needs 1 regen to reach 100 again: 75+30=105 → clamped to 100 ✓). USE_WEAPON (30 cost) also 1-turn. ULTIMATE costs 50 but sets stamina to 0, requiring 4 turns to recover.

4. Entity Turnaround Time Profiles

The table below profiles each entity configuration and their inter-turn TAT for a STRIKE action (most common combat action). Entity speed determines how many total entities share the turn queue; TAT in turns is constant per the stamina model, but the real-time TAT scales with queue depth.

Entity	MAX_STA	STA After STRIKE	TAT (turns)	TAT (ms est.)	Notes
Player (1P)	100	75	1 turn	~5ms	Queue: 1P + N enemies; TAT in turns always 1 for STRIKE
Player (2P)	100	75	1 turn	~5ms	Each player shares queue fairly; same stamina model
Player (4P)	100	75	1 turn	~5ms	More players → longer real-time wait but 1 turn TAT
Enemy (min spd=10)	150	125	1 turn	~5ms	All enemies have same STA model; STRIKE always 1 turn
Enemy (mid spd=20)	150	125	1 turn	~5ms	Speed not used in stamina model; uniform regen
Enemy (max spd=30)	150	125	1 turn	~5ms	All enemies recover in 1 turn after STRIKE
Post-SKIP (player)	100	100	1 turn	~5ms	+STA_SKIP_REGEN brings STA to max; always 1 turn next
Post-ULTIMATE (player)	100	0	4 turns	~20ms	Stamina set to 0; needs 4 regen cycles
Stunned entity	-	-	N/A	3–10s wall	Removed from queue; sleep() for stun_end_time - now

5. Ultimate Ability — Extended Turnaround Impact

The Ultimate Ability is the most scheduling-intensive action in the game. It interacts with three OS-level mechanisms simultaneously: signal delivery (SIGALRM), process suspension (SIGSTOP / SIGCONT), and the stamina scheduler. Below is the full TAT breakdown.

5.1 Timeline of Events

t (rel)	Actor	Event
t = 0	Arbiter	Player action ULTIMATE received via action_ready_cond. apply_action() called.
t = 0+	Arbiter	All alive enemies: stun_active=true, stun_end_time=now+10, state=STUNNED. Player stamina→0.
t = 0+	Arbiter	kill(asp_pid, SIGSTOP) — kernel atomically freezes all ASP threads.
t = 0+	Arbiter	alarm(10) — SIGALRM scheduled for t=10s. ultimate_pause_active=true.
t = 0+	Scheduler	pick_next_actor() enters ultimate-pause branch: only players eligible.
t ≈ 5ms	Scheduler	Player rescheduled (stamina 0+30=30 ≥ STA_MIN_ACT=10). Player attacks.
t ≈ 10ms	Scheduler	Player rescheduled again (stamina after STRIKE: 30-25=5 < MIN → force skip, +40 → 45).
t ≈ 15ms	Scheduler	Player rescheduled (stamina 45+30=75). Player attacks again.
t = 10s	Kernel	SIGALRM delivered to Arbiter. handle_sigalarm() sets ultimate_alarm_fired=1.
t = 10s+	Arbiter	Scheduler loop detects ultimate_alarm_fired. ultimate_pause_active=false.
t = 10s+	Arbiter	kill(asp_pid, SIGCONT) — kernel resumes all ASP threads.
t = 10s+	Scheduler	Normal pick_next_actor() resumes. Enemies stun_end_time expired → recover.

5.2 Player TAT During Ultimate Window

With enemy stamina frozen at its current value and enemies excluded from pick_next_actor(), the player gets uncontested turns. Each turn the player's TAT depends on their stamina:

Turn #	STA Before	Action	STA After	Notes
1	0+30=30	STRIKE	30-25=5	STA after < MIN_ACT(10) → next turn is forced SKIP
2	5+30=35	FORCE SKIP	35+40=75	Forced because STA<10 at start; large regen bonus
3	75+30=100	STRIKE	100-25=75	Back to full; player can act freely again
4	75+30=100	STRIKE	75	Continues until 10s SIGALRM fires

The fix: prior to the bug fix, pick_next_actor() advanced turn_queue_index past the player while scanning stunned enemies, causing it to return -1 and freezing the game. After the fix, the ultimate-pause branch scans independently without modifying the queue index, guaranteeing the player is always found and scheduled.

5.3 Enemy TAT During Stun

Enemies with stun_active=true are skipped by pick_next_actor() and not assigned turns. Their TAT for this forced-skip period is determined entirely by wall-clock time, not turns:

Enemy TAT during ULTIMATE = ULTIMATE_PAUSE_SEC = 10 seconds (wall clock)

After stun expires, pick_next_actor() detects now >= stun_end_time, clears stun_active, adds STA_REGEN to the enemy's stamina, and returns the enemy as the next actor. From the enemy's perspective this is a single forced skip with 10-second latency.

6. Regular Stun Mechanic — Turnaround Analysis

The STUN_DURATION_SEC = 3 mechanic (from EXHAUST or other stunners) follows the same wall-clock model as the Ultimate's 10-second stun, but affects only a single target:

- **Stun initiator TAT:** Normal TAT for EXHAUST (1 turn, ~5ms)
- **Stun victim TAT:** Entire 3-second stun window + scheduler overhead to re-pick them
- **Queue effect:** Other entities continue taking turns during the victim's stun — no global freeze
- **Stamina:** Preserved at pre-stun level. STA_REGEN still applied when victim's stun expires and they are re-selected.

Stun victim TAT = STUN_DURATION_SEC + t_pick ≈ 3.005 seconds

7. Summary & Comparative Analysis

Action / Event	TAT (turns)	TAT (wall time)	Scheduling Impact
STRIKE	1	~5 ms	Minimal; entity recovers in 1 regen cycle
EXHAUST	1	~5 ms	Minimal; but victim loses 3s (stun)
USE_WEAPON	1	~5 ms	Same as STRIKE; cost=30 recovered in 1 cycle
SWAP_IN	1	~5 ms	Low cost, fastest recoverable action
HEAL	1	~5 ms	Negligible TAT; used for stamina preservation
ACQUIRE_ARTIFACT	1	~5 ms	Low cost; blocking on mutex if artifact held
SKIP (voluntary)	1	~5 ms	Fastest possible; +STA_SKIP_REGEN keeps STA full
ULTIMATE (actor)	4	~20 ms	Highest personal TAT; actor starved for 4 turns
ULTIMATE (enemies)	N/A	10,000 ms	Full 10-second process suspension via SIGSTOP
Regular Stun (victim)	N/A	3,005 ms	Target sleeps for stun duration; skips turn
NPC Timeout (auto-	1	3,005 ms	3s wall wait before SKIP applied; scheduler

skip)			continues
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Key Observations

- Most actions have a TAT of exactly 1 turn because STA_REGEN = 30 covers the cost of every common action (STRIKE=25, EXHAUST=20, HEAL=15) within a single regen cycle.
- ULTIMATE is unique in setting actor stamina to zero, requiring $\text{ceil}(100/30) = 4$ turns to recover — the only action with a multi-turn TAT for the actor.
- The stun mechanics (both STUN_DURATION_SEC=3 and ULTIMATE_PAUSE_SEC=10) move turnaround time into the wall-clock domain. The scheduler has no way to shorten these; they are governed by real time via `time(nullptr)` comparisons.
- NPC timeout (3s) is effectively an implicit SKIP with real-time penalty. It only triggers if the ASP fails to submit an action within NPC_TIMEOUT_SEC, which in practice is rare.
- The bug fix for the Ultimate freeze means TAT during the ultimate window is now well-defined: players receive uninterrupted turn assignments until the SIGALRM fires, yielding approximately $\text{floor}(10\text{s} / 5\text{ms}) = \sim 2000$ potential scheduler loops within the 10-second window, bounded in practice by human/GUI input latency.

Chrono Rift — CS 2006 Operating Systems Spring 2026 | Turnaround Time Analysis

Screenshots:

CHRONO RIFT

Enemies Slain: 0 / 10

BATTLE

INVENTORY

ARTIFACTS

ARTIFACT RESOURCES

S

Solar Core

DMG: 95 | Slots: 10

Held by:

Enemy-1

L

Lunar Blade

DMG: 90 | Slots: 10

Held by:

Enemy-2

E

Eclipse Relic

DMG: 60 | Slots: 5

Not Yet Spawned

ACTION LOG

[ARTIFACT] Enemy-2 wields the Lunar Blade!

[ARBITER] HIP spawned (pid 4518)

[ARBITER] ASP spawned (pid 4519)

=== CHRONO RIFT BEGINS ===

Players: 2 | Enemies: 9 | Seed: 240534

[TURN] Hero-1's turn!

[TIMEOUT] Hero-1 timed out auto SKIP

Hero-1 SKIPS stamina restored to 100/100.

[TURN] Enemy-1's turn!

Enemy-1 STRIKES Hero-2 for 13 damage! (HP: 240671/240684)

[TURN] Hero-2's turn!

[TIMEOUT] Hero-2 timed out auto SKIP

Hero-2 SKIPS stamina restored to 100/100.

[TURN] Enemy-2's turn!

Enemy-2 uses Lunar Blade on Hero-2 for 90 damage!

[TURN] Enemy-3's turn!

Enemy-3 STRIKES Hero-2 for 13 damage! (HP: 240568/240684)

[TURN] Enemy-4's turn!

Enemy-4 SKIPS stamina restored to 150/150.

[TURN] Enemy-5's turn!

Enemy-5 STRIKES Hero-2 for 13 damage! (HP: 240555/240684)

[TURN] Enemy-6's turn!

Enemy-6 STRIKES Hero-2 for 13 damage! (HP: 240542/240684)

[TURN] Enemy-7's turn!

Enemy-7 STRIKES Hero-2 for 13 damage! (HP: 240529/240684)

[TURN] Enemy-8's turn!

Enemy-8 STRIKES Hero-2 for 13 damage! (HP: 240516/240684)

[TURN] Enemy-9's turn!

Enemy-9 STRIKES Hero-2 for 13 damage! (HP: 240503/240684)

[TURN] Hero-1's turn!

CHRONO RIFT

Enemies Slain: 0 / 10

BATTLE

INVENTORY

ARTIFACTS

Hero-1

Primary (2/20 slots)

Sp	Sp								
12	12								

LTS (0 Items)

Hero-2

Primary (2/20 slots)

Sp	Sp								
12	12								

LTS (0 Items)

ACTION LOG

[ARTIFACT] Enemy-1 wields the Solar Core!
[ARTIFACT] Enemy-2 wields the Lunar Blade!
[ARBITER] HIP spawned (pid 4518)
[ARBITER] ASP spawned (pid 4519)
=== CHRONO RIFT BEGINS ===
Players: 2 | Enemies: 9 | Seed: 240534
[TURN] Hero-1's turn!
[TIMEOUT] Hero-1 timed out [] auto SKIP
Hero-1 SKIPS [] stamina restored to 100/100.
[TURN] Enemy-1's turn!
Enemy-1 STRIKES Hero-2 for 13 damage! (HP: 240671/240684)
[TURN] Hero-2's turn!



Chrono Rift — Turnaround Time Analysis

Page 9